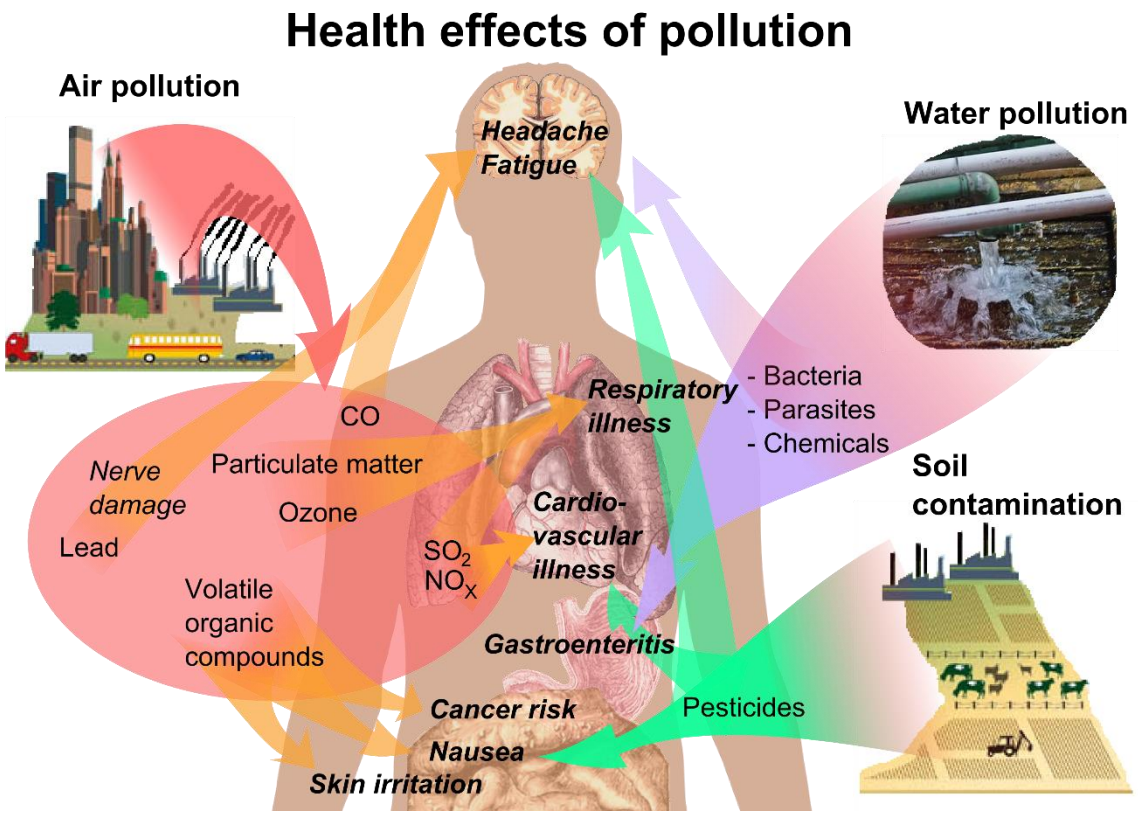
Environmental Pollutants

Debasree Purkayastha
School of Infrastructure, IIT Bhubaneswar
Email id: debasreep@iitbbs.ac.in

What is Environmental Pollution?

- Environmental pollution is unwarranted disposal of mass or energy into earth's natural resource pool such as water, land, or air that results in long- or short-term detriment to the atmosphere and its ecological health to negatively impact the living beings and their life both quantitatively and qualitatively.



Source: 1. Yadav, Poonam, Kalidindi Usha, and Bhupinder Singh. "Air pollution mitigation and global dimming: a challenge to agriculture under changing climate." In Climate change and crop stress, pp. 271-298. Academic Press, 2022. 2. Bing Search engine, CCBY

What is Environmental Pollutants?

- Environmental pollutants are substances that, once introduced into the environment, alter its natural composition and pose direct or indirect threats to human survival and ecological well-being.

Source: 1. Sciencedirect, https://www.sciencedirect.com/topics/chemistry/environmental-pollutant 2. Image: iStock, https://www.istockphoto.com/photos/air-pollution-concept, Accessed on: 11 January, 2025

Types of Pollutants

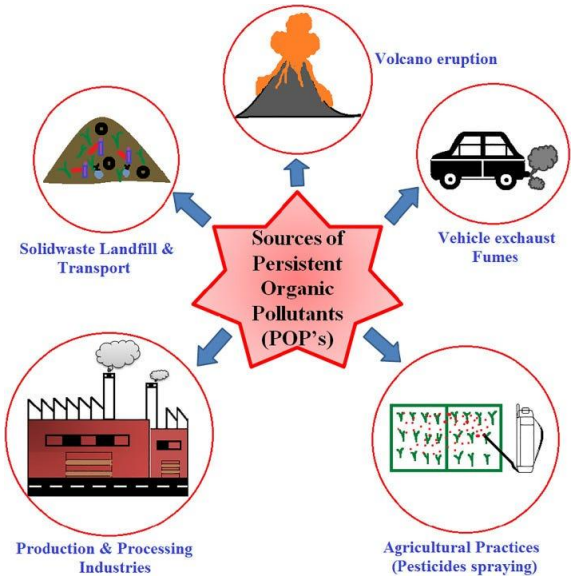
- **Persistent organic pollutants (POPs)** are toxic chemicals that adversely affect human health and the environment around the world. Because they can be transported by wind and water, most POPs generated in one country can and do affect people and wildlife far from where they are used and released. They persist for long periods of time in the environment and can accumulate and pass from one species to the next through the food chain.

- **Non-persistent organic pollutants (NOPs)** are substances that can cause damage to organisms when added in excessive amounts to the environment but is decomposed or degraded by natural biological process and removed from the environment relatively quickly.

Basic type of pollutants

Persistent Pollutant (POPs)

Non-Persistent Pollutant (NOPs)



Source: 1. Yadav, Poonam, Kalidindi Usha, and Bhupinder Singh. "Air pollution mitigation and global dimming: a challenge to agriculture under changing climate." In Climate change and crop stress, pp. 271-298. Academic Press, 2022. 2. Deng, Man, et al. "Are human exposure assessment the same for non-persistent organic chemicals?-from the lens of urinary variability and predictability." Science of The Total Environment 868 (2023): 161542. 3. Bing search, CCBY

DDT Dilemma

- DDT is likely one of the most famous and controversial pesticides ever made. An estimated 4 billion pounds of this inexpensive and historically effective chemical have been produced and applied worldwide since 1940. In the United States, DDT was used extensively on agricultural crops, particularly cotton, from 1945 to 1972. DDT was also used to protect soldiers from insect-borne diseases such as malaria and typhus during World War II, and it remains a valuable public health tool in parts of the tropics. The heavy use of this highly persistent chemical, however, led to widespread environmental contamination and the accumulation of DDT in humans and wildlife - a phenomenon brought to public attention by Rachel Carson in her 1962 book, Silent Spring. A wealth of scientific laboratory and field data have now confirmed research from the 1960s that suggested, among other effects, that high levels of DDE (a metabolite of DDT) in certain birds of prey caused their eggshells to thin so dramatically they could not produce live offspring.

- One bird species especially sensitive to DDE was the bald eagle. Public concern about the eagles' decline and the possibility of other long-term harmful effects of DDT exposure to both humans and wildlife prompted the Environmental Protection Agency (EPA) to cancel the registration of DDT in 1972. The bald eagle has since experienced one of the most dramatic species recoveries in our history.

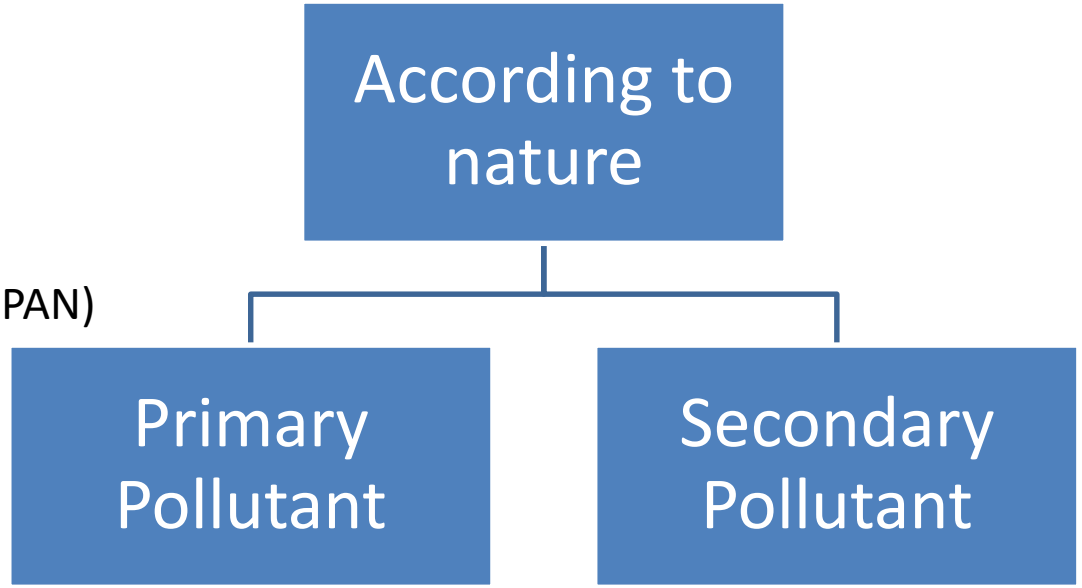
Source: 1. Persistent pollutant, https://www.epa.gov/international-cooperation/persistent-organic-pollutants-global-issue-global-response, Accessed on: 12 January, 2024

Types of Pollutants

- **Primary pollutants** are the ones that are discharged directly into the atmosphere from natural or artificial sources. (CO, NOx, dust)

(Ozone, Smog, PAN)

- **Secondary pollutants** are produced by chemical reactions or physical interactions between primary pollutants or between primary pollutants and other atmospheric components. (Ozone, Smog, PAN)



Source: 1. Sciencedirect, https://www.sciencedirect.com/topics/chemistry/environmental-pollutant 2. Image: iStock, https://www.istockphoto.com/photos/air-pollution-concept, Accessed on: 11 January, 2025

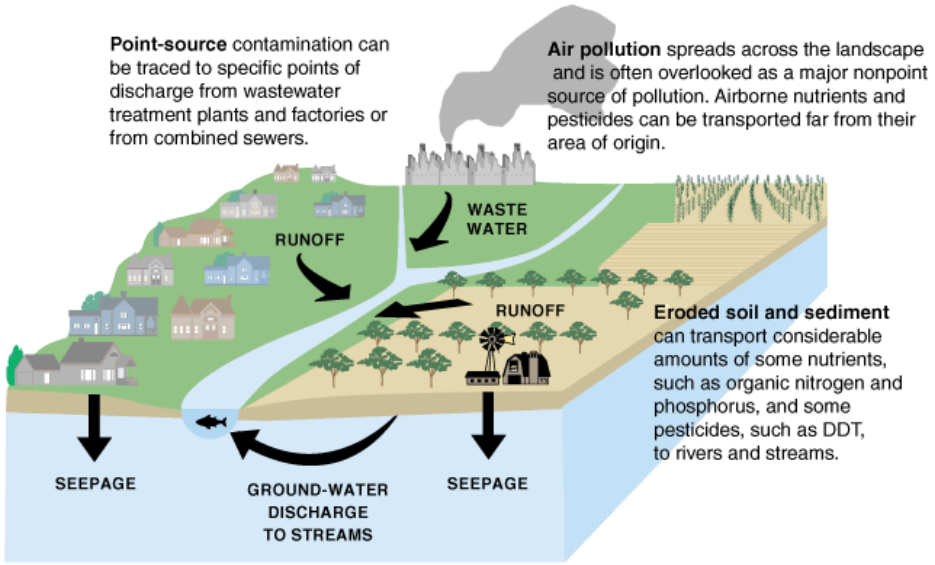
Types of Pollutants

- The U.S. Environmental Protection Agency (EPA) defines **Point Source** pollution as "any single identifiable source of pollution from which pollutants are discharged, such as a pipe, ditch, ship or factory smokestack."
- **Non-Point Source** pollution is caused by rainfall or snowmelt moving over and through the ground. As the runoff moves, it picks up and carries away natural and human-made pollutants, finally depositing them into lakes, rivers, wetlands, coastal waters and ground waters.

According to origin

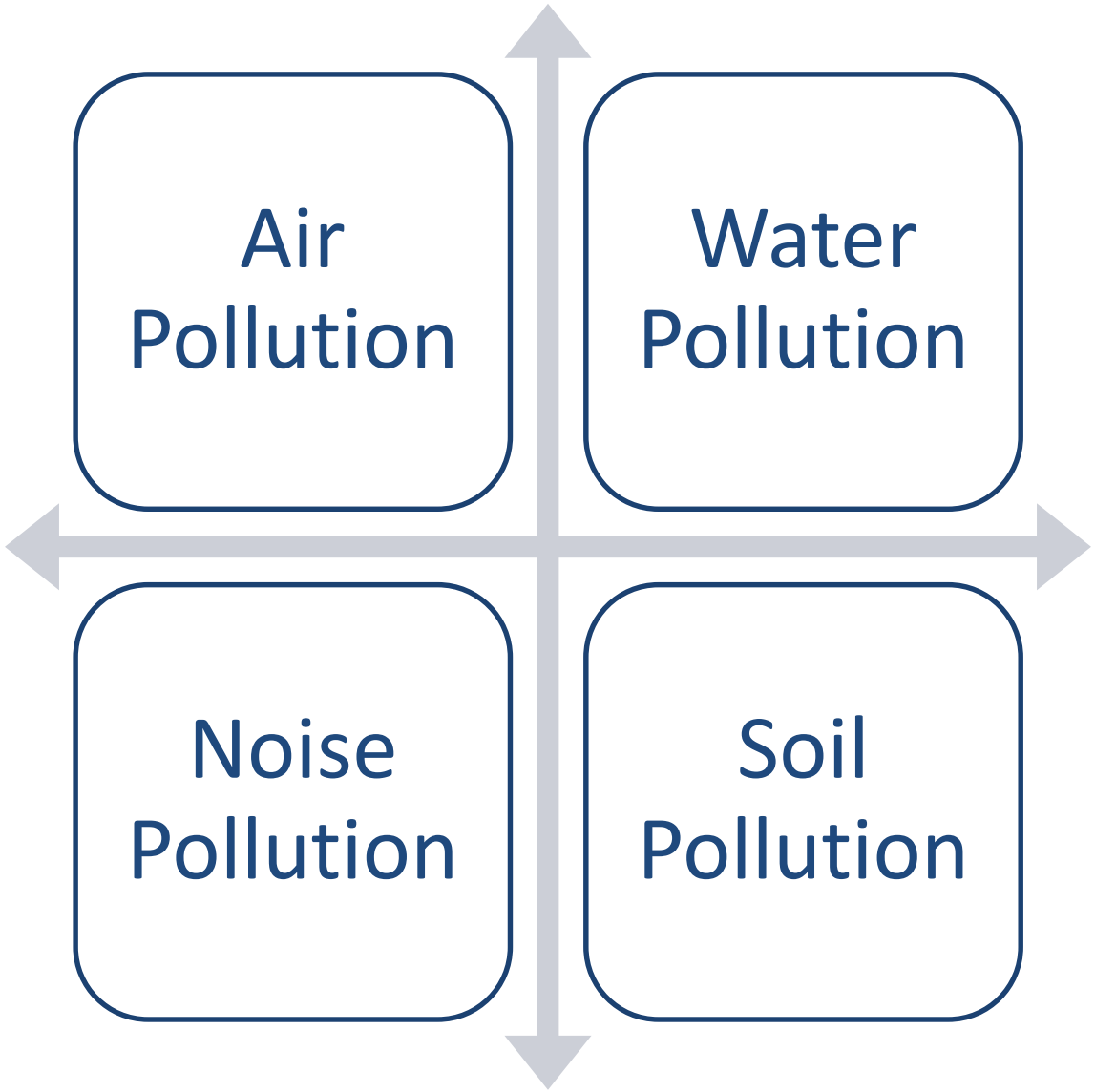
Point Source

Non-Point Source



Source: 1. Point Source, https://oceanservice.noaa.gov/education/tutorial_pollution/03pointsource.html 2. Non point source, https://www.epa.gov/nps/basic-information-about-nonpoint-source-nps-pollution 3. Bing search, CCBY Accessed on: 11 January, 2025

Types of Environmental Pollution



Water Pollution

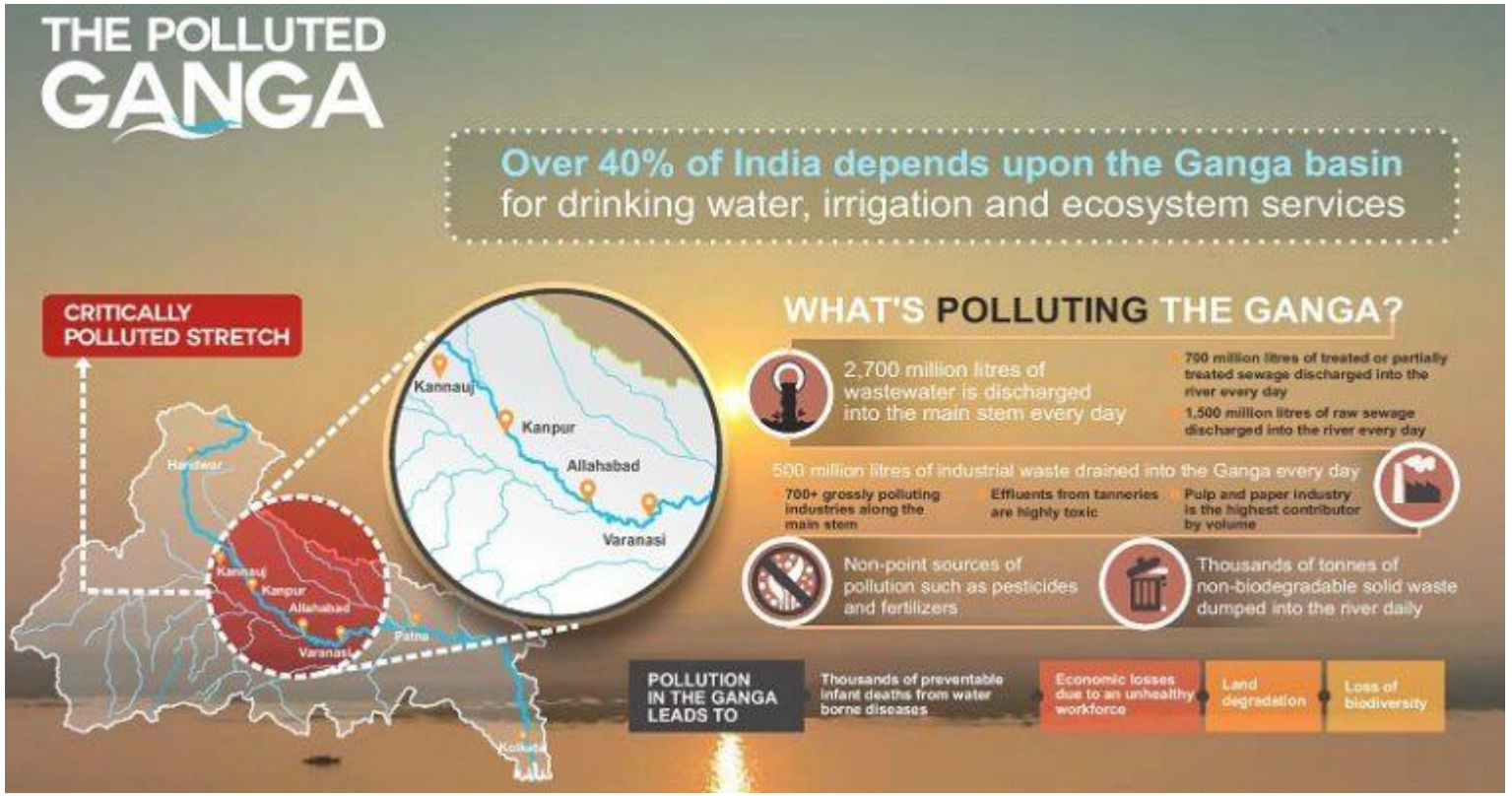
- **Water Pollution** can be defined as alteration in physical, chemical, or biological characteristics of water through natural or human activities and making it unsuitable for its designated use.
- Physical characteristic refers to colour, taste, turbidity and smell of water.
- Chemical characteristic refers to- pH, TDS, DO, COD, BOD.
- Biological characteristic refers to- presence of biological components i.e. bacteria, fungi, etc.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Bing Search, CCBY

Water Pollution

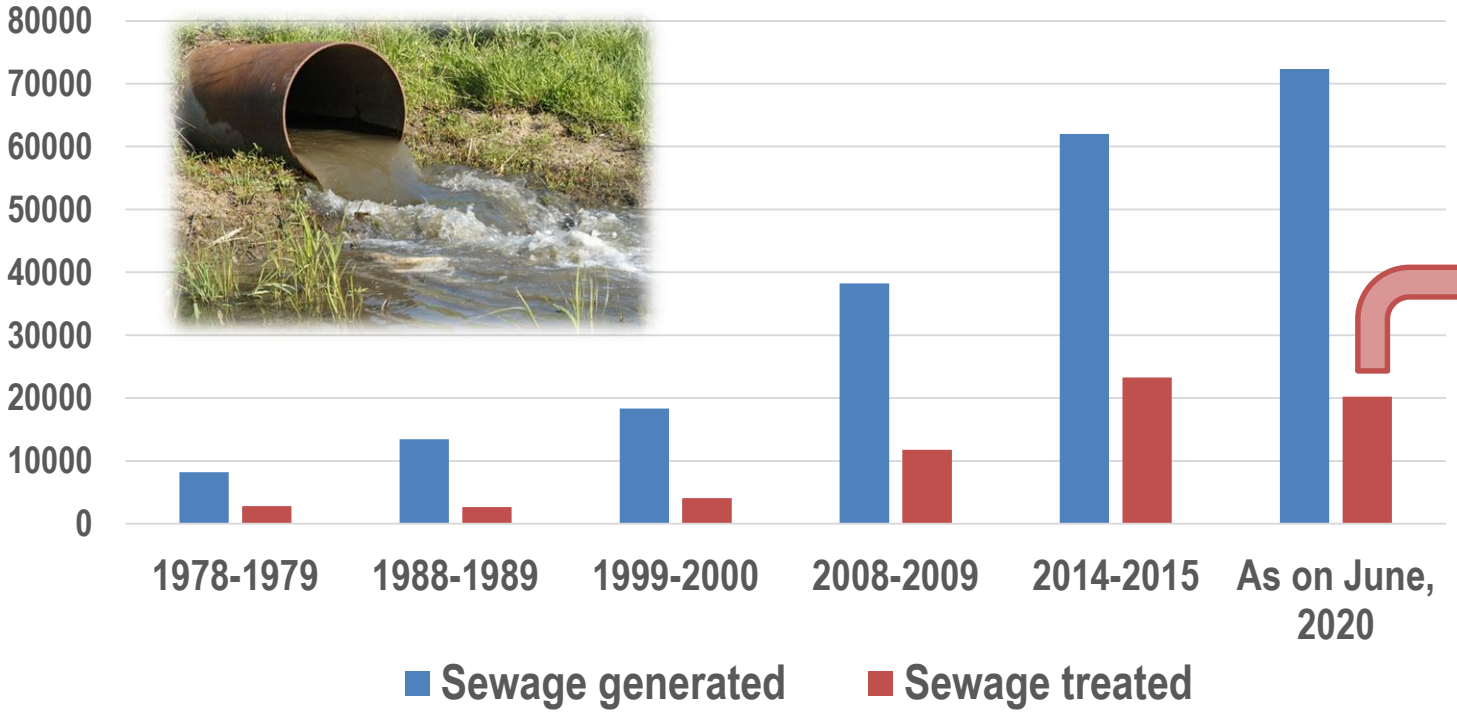
The sources of water pollution can be classified as POLLUTION:

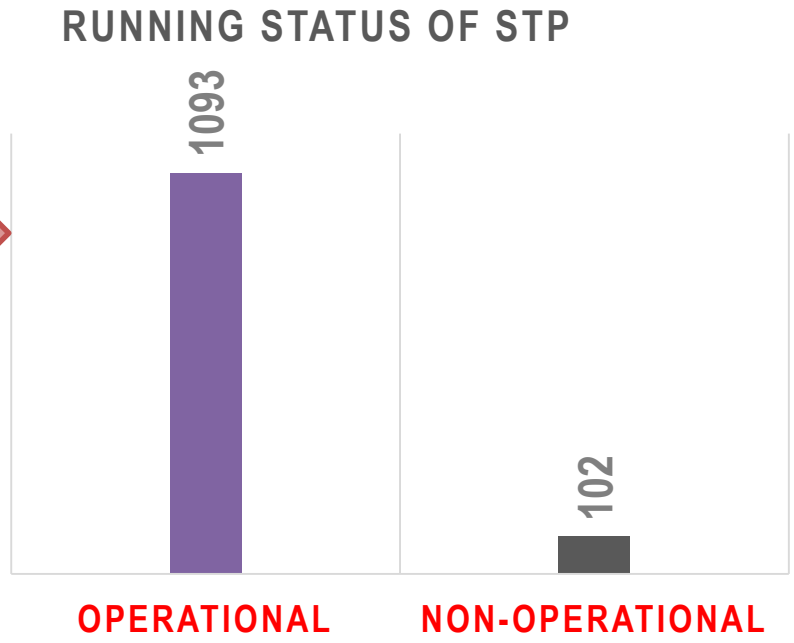
- Municipal Waste Water
- Industrial Pollution- wastes from industries, metals, etc.
- Inorganic Pollutants- fine particles of heavy metals, chlorides,
- Organic Pollutants- oil, phenols, organic acids like HNO_3 etc.
- Agricultural Wastes- Chemical fertilizers, pesticides
- Marine Pollution – Grease, oils, detergents,
- Thermal pollution -Thermal plants and nuclear plants



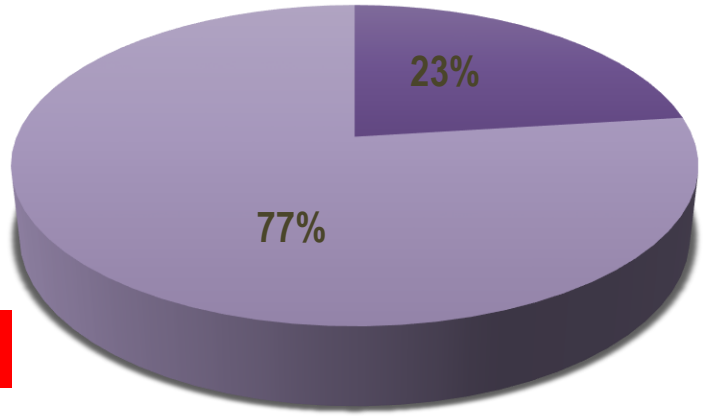
Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image source: Causes of Ganges River Pollution, https://knowledgeofindia.com/causes-ganges-river-pollution-case-study/

Scenario of Indian Sewage treatment (in MLD)





STP: SEWAGE TREATMENT PLANT



POOR OPERATION & MAINTENENCE

LOAD VARIATION

Reference: Central Pollution Control Board, National Inventory of Sewage Treatment Plants, March 2021

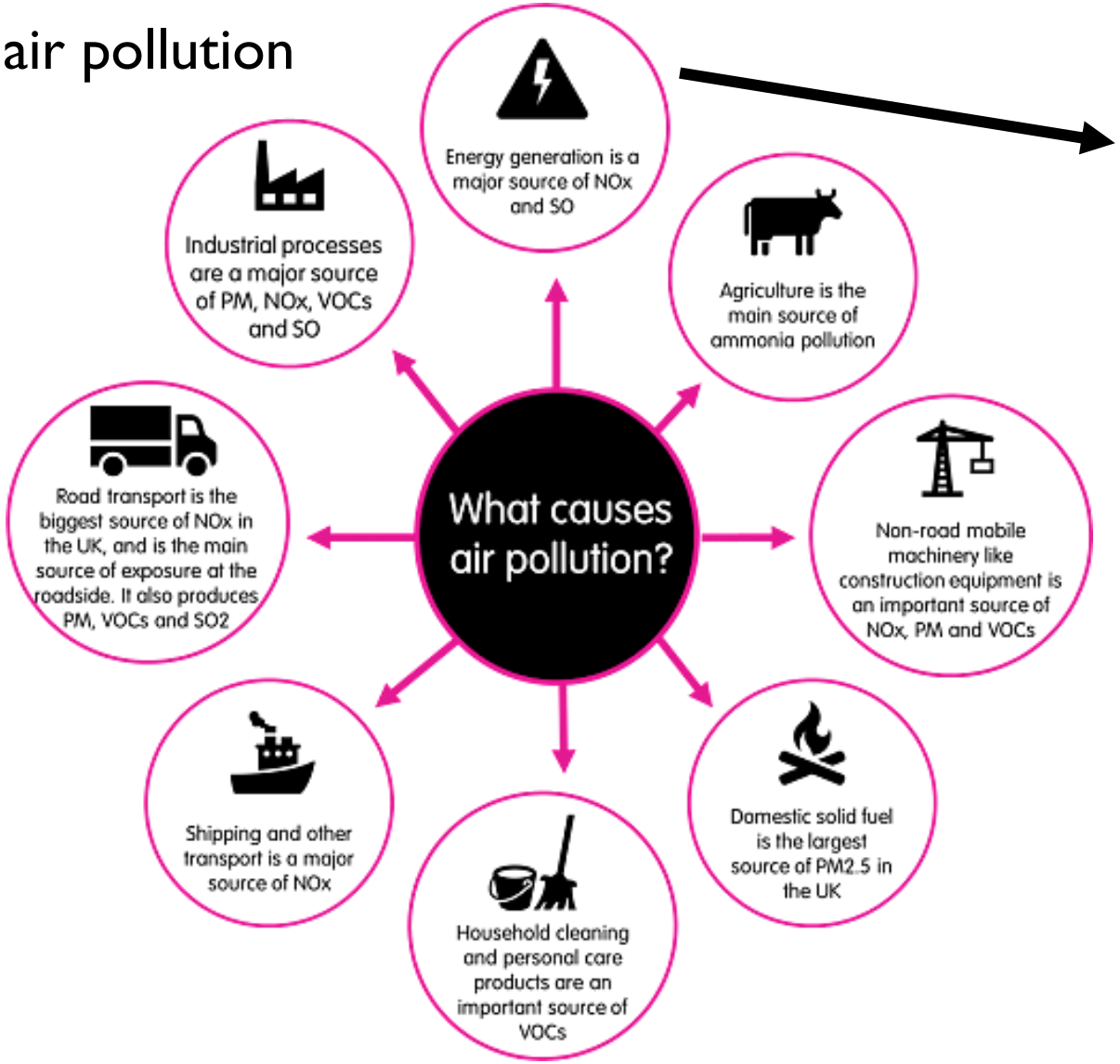
Air Pollution

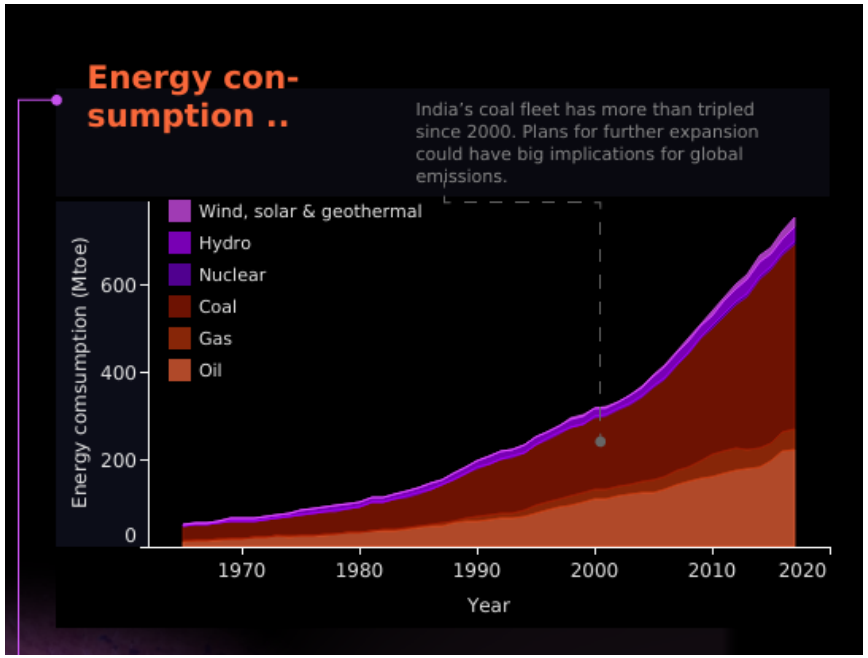
- Air pollution is defined as the release of pollutants into the atmosphere that are harmful to human health and the environment as a whole.
- According to the World Health Organization (WHO), air pollution kills approximately seven million people per year around the world.
- Nine out of ten people now breathe air that exceeds the WHO's pollution guideline levels.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Bing Search, CCBY

Air Pollution

- Causes of air pollution





Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Image: https://www.carbonbrief.org/the-carbon-brief-profile-india

Major Air Pollutants

Carbon monoxide

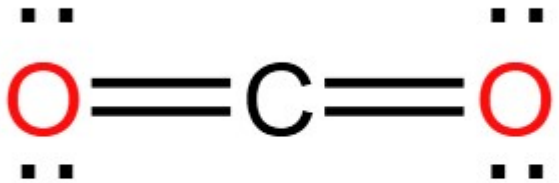
- It is a colorless, odorless gas created by incomplete combustion of carbon-based fuels such as gasoline, diesel, and wood, as well as the combustion of natural and manmade items like cigarettes.
- It reduces the amount of oxygen in our bloodstream.
- It can make us confused and tired by slowing our reflexes and making us disoriented.

Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Bing Search, CCBY

Major Air Pollutants

Carbon dioxide (CO_2)

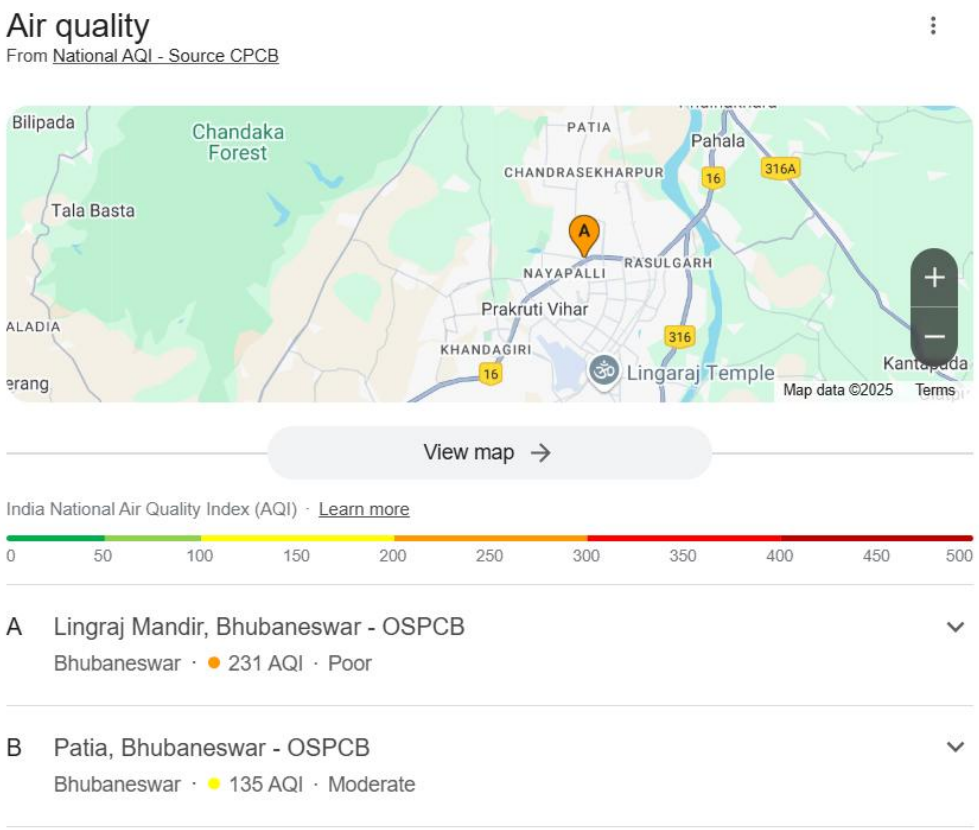
- It is the primary greenhouse gas emitted as a result of human activities such as coal, oil, and natural gas combustion.
- Volcanoes, hot springs, and geysers are natural sources of CO_2, and it is liberated from carbonate rocks by dissolving in water and acids.
- CO_2 is a suffocating gas (asphyxia: a condition arising when the body is deprived of oxygen, causing unconsciousness or death.).
- Even in the presence of sufficient oxygen, concentrations of 7% can produce asphyxia, which manifests as dizziness, headache, and unconsciousness.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Bing Search, CCBY

Air Pollution

- Air quality is measured using air quality indicators like the " Air Quality Index". PM 2.5 and PM 10 are the most often used air quality indicators.
- The particles are measured in micrograms per cubic meter.
- PM 2.5 denotes the concentration of microscopic particles with a diameter of fewer than 2.5 microns, while PM 10 denotes the concentration of particles with a diameter of less than 10 microns.
- India also monitors two more pollutants: lead and ammonia.
- A number of less than 50 on the AQI scale is deemed safe.



Source: 1. Howard S. Peavy et al., Environmental engineering, McGraw Hill Education (1985), 2. Google Search, AQI of Bhubaneswar, Accessed on 12 Jan, 2025

Soil Pollution

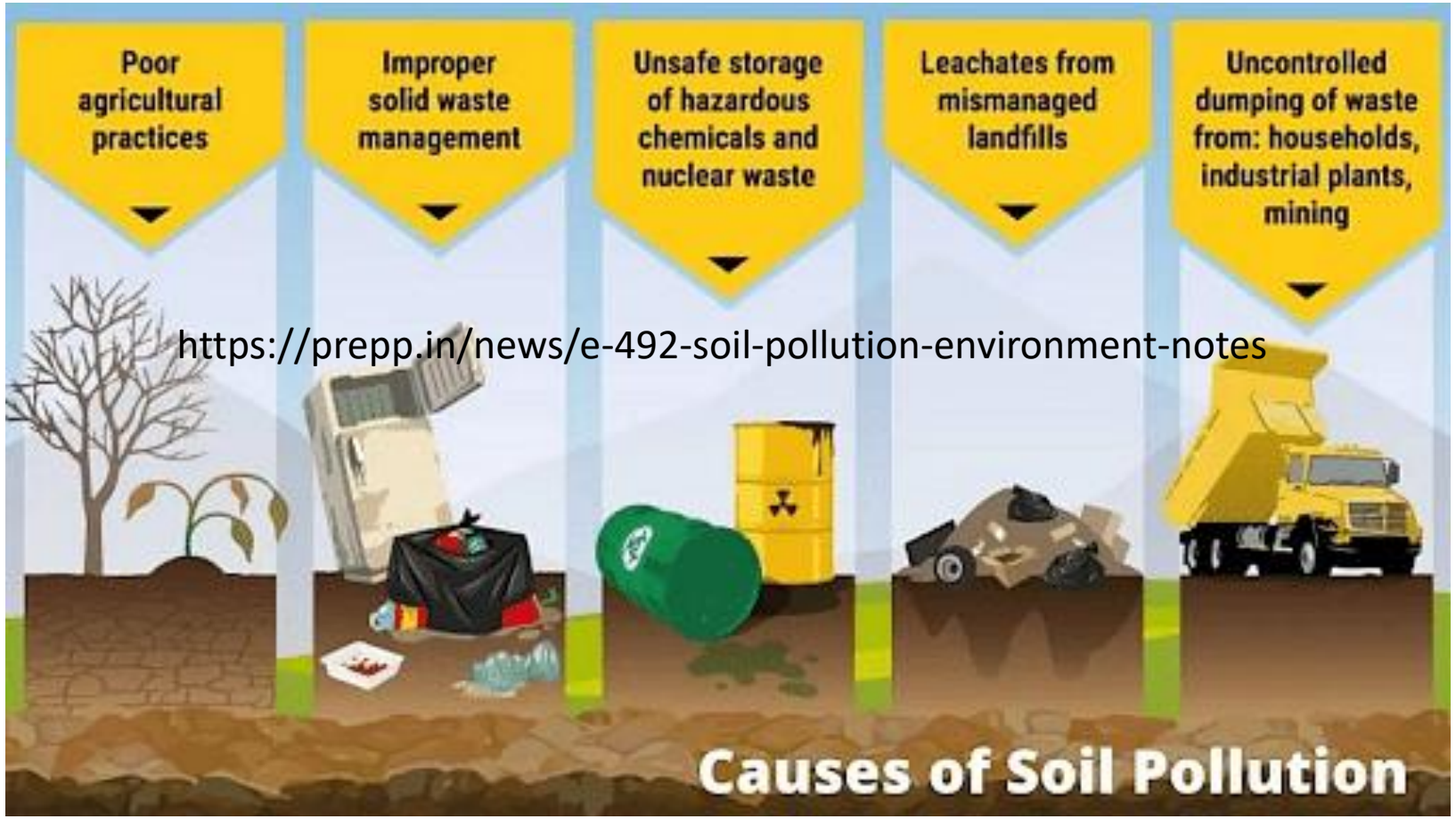
- The presence of harmful compounds (also known as pollutants or contaminants) in the soil at quantities high enough to endanger human health and/or the ecosystem is referred to as soil pollution.

- According to the Food and Agricultural Organization (FAO) of the United Nations, soil pollution is a threat that affects the entire world but is particularly serious in regions like Europe, Eurasia, Asia, and North Africa.

Source: 1. FAO, Report sounds alarm on soil pollution, https://www.fao.org/newsroom/detail/Report-sounds-alarm-on-soil-pollution/es, Accessed on: 12 January, 2025

Soil Pollution

- Causes of Soil pollution



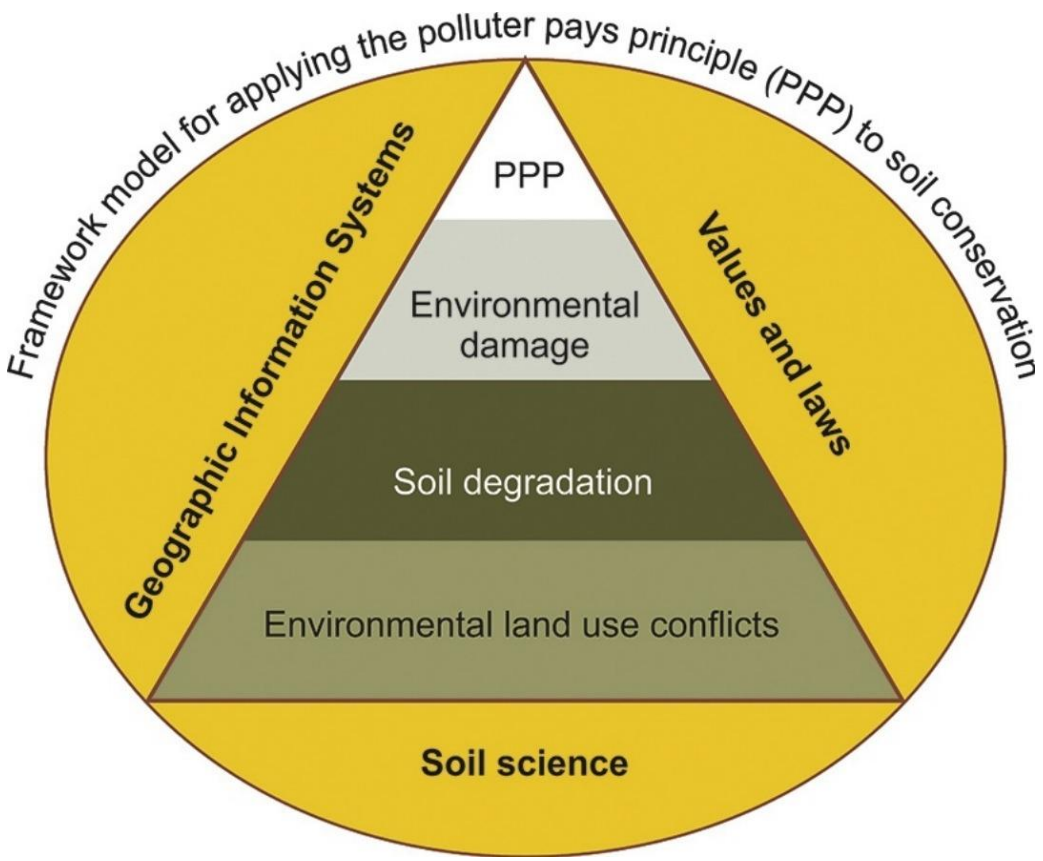
https://prepp.in/news/e-492-soil-pollution-environment-notes

Source: 1. T.V. Ramachandra, NPTEL Lecture notes on EIA, 2. Soil Pollution, https://prepp.in/news/e-492-soil-pollution-environment-notes

Soil Pollution

Measures to control Soil pollution

- Reduced Use of Chemical Fertilizers
- Promotion of Reforestation and Afforestation
- Recycle and Reuse Products



Source: 1. Polluter-Pays Principle, https://www.sciencedirect.com/topics/earth-and-planetary-sciences/ 2. Valera, C. A., et al. "A legal framework with scientific basis for applying the 'polluter pays principle'to soil conservation in rural watersheds in Brazil." Land use policy 66 (2017): 61-71.

Noise Pollution

- Any undesired or irritating sound that impacts the health and well-being of humans and other organisms is referred to as noise pollution.
- According to a WHO estimate from 2007, over 6% of Indians suffer from hearing loss. Noise-Induced Hearing Loss is caused by prolonged exposure to noise levels above 60 decibels (NIHL).
- Deafness, heart issues, sleep disorders, and cognitive impairment in youngsters are a few of the major health consequences.
- Noise Pollution can be controlled by designing quieter machines, using acoustic enclosures, and vibration isolators, increasing the distance of the transmission and using noise-canceling headphones, earplugs, and more.

Source: 1.Noise Pollution, https://prepp.in/news/e-492-noise-pollution-environment-notes, 2. Bing Search, CCBY, Accessed on: 12 January, 2025

Source: BBC News, Five fires still burn across LA as fierce winds threaten more destruction, https://www.youtube.com/watch?v=DwO9HajdTc4, Accessed on: 12 January, 2025

Thank you

For any questions, please feel free to reach out to me.